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Semester VI (B.Tech)

Er. No.....
Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1(Even Semester 2023)

18B14CI842 – MOBILE COMPUTING

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has five questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

	Marks	CO No.
Q1. Elucidate the characteristics of mobile computing. What are the major limitations of mobile computing?	[03]	CO2
Q2. Describe the categories of MAC Protocols? Compare different fixed assignment MAC protocols.	[03]	CO3
Q3. Draw TDMA frame, slots and bursts in terms of time duration, fields and size of each field. How many total channels are there in uplink and downlink?	[03]	CO4
Q4. What are the important functions of Network Switching Subsystem? Explain the various functional units of NSS with block diagram.	[03]	CO3
Q5. Illustrate the key features and disadvantages of second and third generations of mobile communication systems.	[03]	CO2

Semester VI (B.Tech)

Er. No.....
Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1(Even Semester 2023)
18B11CI612 – SOFTWARE ENGINEERING

Maximum Marks: 15

Maximum Duration: 1 Hour

Notes:

1. This question paper has 03 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

Q1.

Consider the following requirements for software to be developed for controlling a chemical plant. The chemical plant has a number of emergency conditions. When any of the emergency conditions occurs, some specified actions should be taken. The different emergency conditions and the corresponding actions that need to be taken are as follows:

- I. If the temperature of the chemical tank is high, then the water shower should be turned ON.
- II. If the temperature of the chemical tank falls below T_1^0 C, then the heater should be turned ON and the water shower should be turned OFF.
- III. If the temperature of the chemical tank exceeds T_1^0 C, then the heater should be turned OFF.
- IV. If the temperature of the chemical tank exceeds T_2^0 C the alarm bell should be sounded.
- V. If the pressure in the tank rises above P_1 , then alarm bell should be sounded.
- VI. If the pressure of the tank goes down, then alarm bell should be OFF.

Identify the problems in the above requirement statements if any.

Marks
[05]

CO No.
CO2

Q2. (a)

Adaptive Software Development (ASD) is a technique for building of complex software systems. With of help of suitable diagram, explain how human collaboration and team self-organization of ASD help in complex software development.

[02]

CO1

- (b) With the help of suitable diagram, explain the different phases of Extreme Programming approach. [03]

(01)

Q3. Consider the following simplified case study "University Web kiosk system" for providing the services to their students and employees:

(03)

The students of the University can check the list of registered subjects, results of previous semesters, personal information, and subject wise attendance status of registered subjects, status of fees deposited, and the names of concerned faculty members for the registered subjects. The employees of the university can be faculty members or administrative staff members. Faculty members can check the list of registered students in their concern courses, personal information of students registered in their concern courses, enter the date wise attendance, enter the marks obtained in different examinations, and apply for a leave. The administrative staff provides the login information to students and employees, prepare the results of students, apply for a leave.

- (a) Identify the actor(s) for use case model [01]
(b) List possible use cases from above case study [01]
(c) Draw the use case diagram for the above mentioned case study. [03]

Jaypee University of Engineering & Technology, Guna**T-1(Even Semester 2023)****18B11CI414 – COMPUTER ORGANISATION AND ARCHITECTURE**

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 03 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

- | | Marks | CO No. |
|---|-------|--------|
| Q1. (a) Consider $(46)_x = (y6)_8$. Find the all possible values of x and y. | [02] | CO2 |
| (b) Represent smallest and largest 4-bit fractional numbers into IEEE 754 single precision format. | [02] | |
| Q2. (a) A computer system executes a program of 10,000 instructions using clock rate of 3 GHz and CPI (cycles per instructions) of 1.5. Find the execution time of the program in nano seconds. Suppose you are required to reduce the execution time to 80% which leads to an increase of 20% in the CPI. What clock rate will be required to get this time reduction? | [02] | CO3 |
| (b) Suppose your computer monitor consumes power of 50 watt per hour, CPU consumes 60 watt per hour and all other hardware components consumes total of 40 watt per hour. How much amount you have to pay monthly @ Rs 5 per unit if computer system is used 12 hours/day. | [02] | |
| Q3. (a) Illustrate the design of full adder using 4:1 multiplexers only. Use initial carry (C_{in}) as input line. | [02] | CO4 |
| (b) Sketch the design of 2-bit, four source shared bus system using tri-state buffers which performs following micro-operations given in the table below: | [02] | |

Control lines	Micro-operations
$\overline{S_1} \overline{S_0}$	$R \leftarrow A$
$\overline{S_1} S_0$	$R \leftarrow B$
$S_1 \overline{S_0}$	$R \leftarrow C$
$S_1 S_0$	$R \leftarrow D$

- (c) Obtain the Boolean expressions and then sketch the design of gate level logic circuit for a seven segment which displays 'O' and 'E' for odd and even octal number inputs respectively. [03]

Semester VI (B.Tech)

Er. No.....

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna**T-1(Even Semester 2023)****18B14CI642 – Computer Graphics**

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 04 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

		Marks	CO No.
Q1.	Describe printing process of a character impact printer (dot-matrix printer). How printing quality is determined of such printers? How light and dark areas are generated in a print by such printers.	[03]	CO1
Q2. ✓	Explain Bresenham's line drawing algorithm for slope < 1 giving its advantages. Design a similar algorithm to draw a line parallel to X-axis from point (x_1, y_1) to (x_2, y_2) .	[05]	CO2
Q3. ✓	Develop a flood-fill algorithm to fill the interior of any specified area.	[05]	CO3
Q4. ✓	Suppose you have a system with an 8-inch by 10-inch video monitor that can display 100 pixels per inch. If memory is organized in one-byte words, the starting frame-buffer address is 0, and each pixel is assigned one byte of storage, what is the frame-buffer address of the pixel with screen coordinates (x, y) .	[02]	CO2